Apple Stock Price Prediction (7 Days)

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

In [2]:
```
# Load or simulate data
# (For real use: download using yfinance)
# Simulated example below:
dates = pd.date_range(end=pd.Timestamp.today(), periods=126, freq='B')
np.random.seed(0)
prices = np.cumsum(np.random.normal(loc=0.5, scale=2, size=len(dates))) + 175

data = pd.DataFrame(data={'Close': prices}, index=dates)
data['Prediction'] = data[['Close']].shift(-7)
```

In [3]:
```
# Prepare training data
X = np.array(data.drop(['Prediction'], axis=1))[:-7]
y = np.array(data['Prediction'])[:-7]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=
```

In [4]:
```
# Train model
model = LinearRegression()
model.fit(X_train, y_train)
```

Out[4]:
```
▾ LinearRegression
LinearRegression()
```

In [5]:
```
# Predict future prices
X_future = data.drop(['Prediction'], axis=1)[-7:]
forecast = model.predict(X_future)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py:457: UserWarning: X has fe
ature names, but LinearRegression was fitted without feature names
  warnings.warn(
```

In [6]:
```
# Visualize
forecast_dates = data.index[-7:]
forecast_df = pd.DataFrame({'Date': forecast_dates, 'Predicted Close': forecast})
forecast_df.set_index('Date', inplace=True)

plt.figure(figsize=(10,5))
plt.plot(data.index[-30:], data['Close'][-30:], label='Actual Price')
plt.plot(forecast_df.index, forecast_df['Predicted Close'], label='Predicted Price', l
plt.xlabel('Date')
plt.ylabel('Price ($)')
plt.title('Apple Stock Price Prediction (7 Days)')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

Apple Stock Price Prediction (7 Days)
Actual Price
Predicted Price
Price ($)
280
270
260
250
240
230
2025-05-01
2025-05-08
2025-05-15
2025-05-22
2025-06-01
2025-06-08
Date